

Project Template: Local LLM Performance & Prompt Comparison

Exploring Local Inference with Ollama

Project Title	[Local LLM Performance & Prompt Comparison]
Student Name	[Meghana Sadula]
Cohort / Batch	[Batch 1]
Date of Submission	[25-05-2026]
Submitted To	[Teja Sai Chenammasetti]

1. Project Objective

This project involves setting up and running a local Large Language Model (LLM) using Ollama. You will test four different prompt styles on a single real-world task to evaluate output quality, generation speed, and the specific capabilities or limitations of running models locally on your hardware.

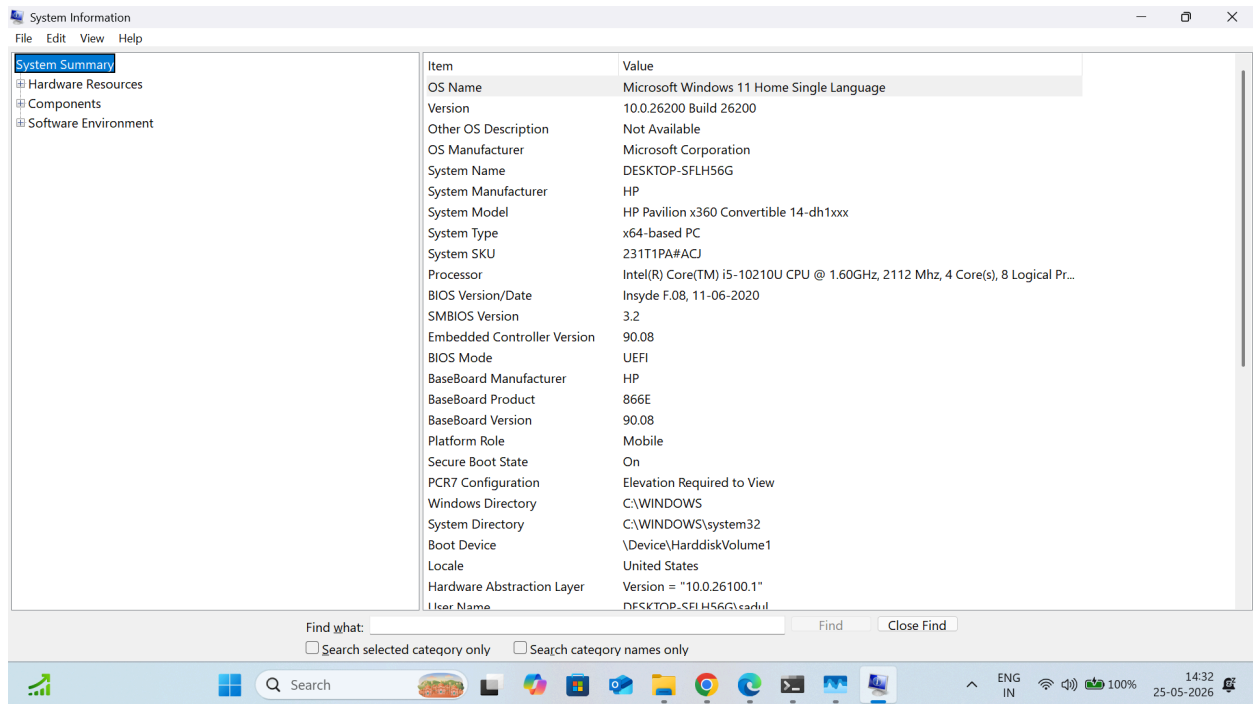
2. System Details

Document the hardware you are using to run the model.

- **Laptop / Desktop Model:** [HP Pavilion x360 Convertible 14-dh1xxx]
- **Processor / Chip (e.g., M1, Intel i7, Ryzen 5):** [Intel(R) Core(TM) i5-10210U CPU @ 1.60GHz]
- **RAM (e.g., 8GB, 16GB, 32GB):** [8GB]
- **Operating System:** [OS Name: Microsoft Windows 11 Home Single Language
OS Version: 10.0.26200 N/A Build 26200]
- **GPU Available or Not:** [Intel(R) UHD Graphics]
- **Hardware Performance:** (Did the model run smoothly / slowly / with issues?): [
Hardware Performance: Slightly slow during loading, Lagging sometimes, Crashed

occasionally, High RAM usage, Memory Issue.]

Screenshot of system details (About This Mac / System Info):



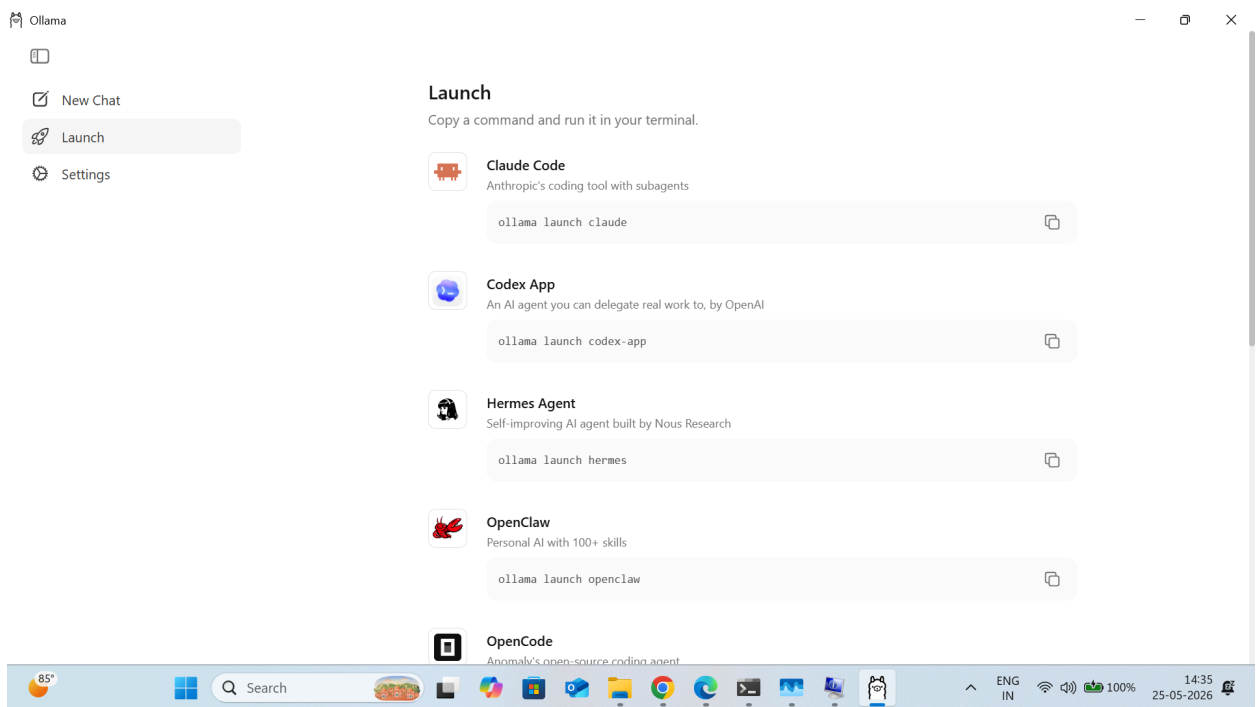
[Paste screenshot here]

3. Ollama Setup

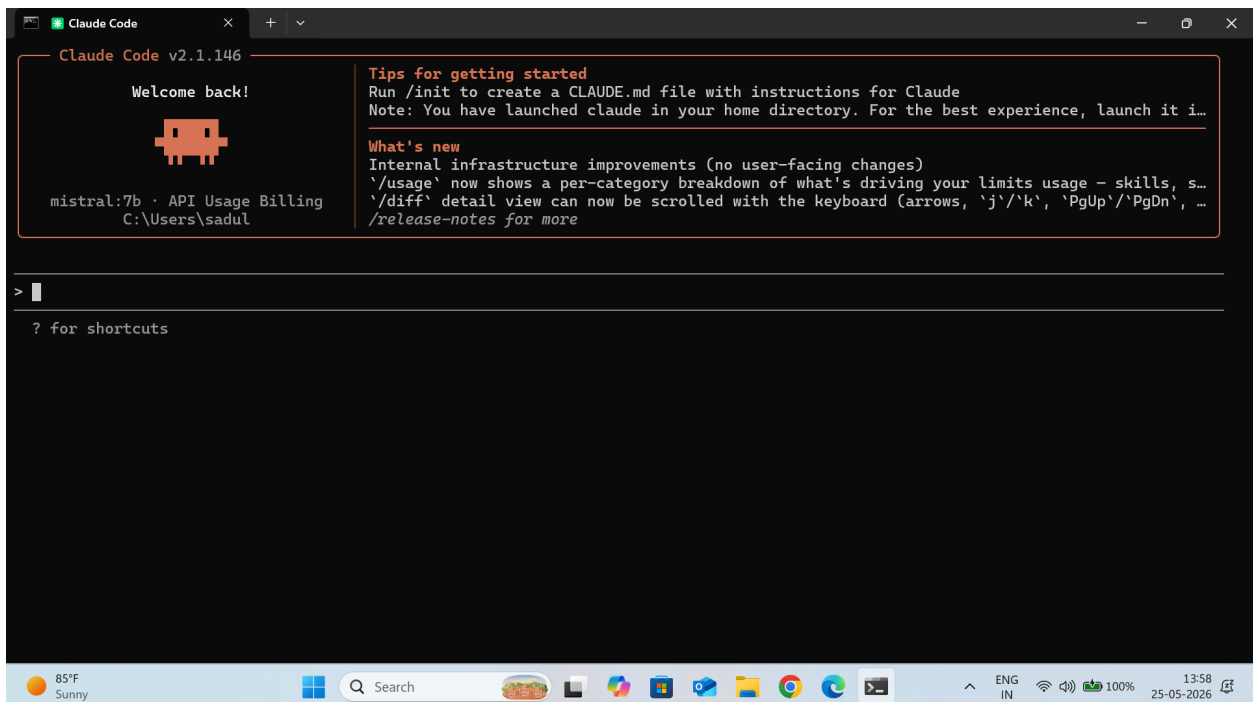
Confirm the installation and initialization of your local environment.

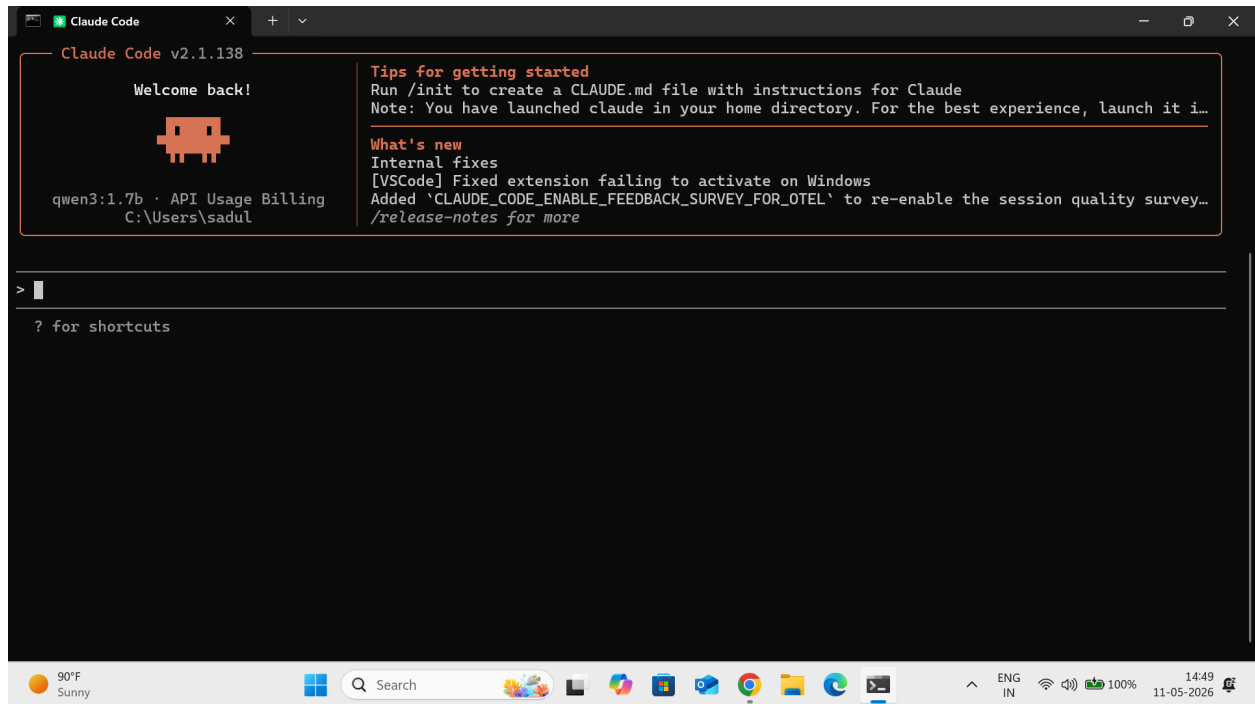
- **Was Ollama installed successfully?:** [Yes]
- **Which model did you pull? (e.g., llama3, mistral, phi3):** [All the above with qwen model]
- **Did the model run successfully?:** [Yes with some issues]
- **Any setup issues faced?:** [Yes, RAM Issue, Memory Issue]

Screenshot of Ollama installation / app / terminal view:



Screenshot showing model pulled or running:





4. Chosen Real-World Task

Describe the specific problem or task you are asking the LLM to solve.

- **Task Chosen:** [Creating a professional LinkedIn post for an AI Engineer.]
- **Why this task was chosen:** [LinkedIn content creation is a real-world activity commonly performed by professionals, marketers, and AI engineers. It tests creativity, formatting, clarity, and instruction-following abilities of the model.]
- **Real-world use case:** [Businesses and professionals use AI tools to generate social media content quickly for branding, marketing, and engagement.]

5. Local Model Used

- **Model Name:** [Qwen3:1.7b]
- **Why this model was selected:** [Qwen3:1.7b is a lightweight small language model that can run efficiently on normal laptops with limited RAM while still generating good-quality responses for prompt testing and experimentation.]
- **Was this model suitable for your system? (Size vs. Speed):** [Yes. The model was

lightweight, responded quickly, and worked smoothly on the local system without requiring high-end GPU hardware.]

6. Prompt Testing Section

Test the same task using four different prompting strategies.

Prompt Version 1 — Zero-shot

Prompt Used: [Write a LinkedIn post about AI Engineering.]

Output Received: [The model generated a short LinkedIn post explaining AI engineering, key skills included in AI, future opportunities, and demand of AI technologies.

AI Engineering: Pioneering Innovation in a Rapidly Evolving Landscape

AI Engineering is at the forefront of transforming industries, blending technical expertise with real-world impact. Key skills include machine learning, data engineering, cloud computing, and collaboration across disciplines. It drives innovations in healthcare, finance, manufacturing, and climate tech, enabling solutions like predictive analytics, autonomous systems, and energy optimization.

The demand for AI Engineers is soaring, with exponential growth in demand. Whether you're a developer, data scientist, or product leader, this field offers unique opportunities to shape the future of technology. Resources like courses, books, and communities support continuous learning.

🌟 #AIEngineering #TechInnovation #FutureOfTech.]

Observation:

[

- Quick response
- Generic output
- Less engaging
- Minimal formatting

I observed that the original LinkedIn post detailed AI Engineering's role in driving innovation across industries, emphasized key skills (machine learning, data engineering, cloud computing), highlighted impact areas (healthcare, finance,

manufacturing, climate tech), and provided resources for learning. The summary condensed this into a concise version while retaining the core message: the demand for AI Engineers, their role in shaping technology, and opportunities for professionals. The assistant focused on preserving critical elements like skills, impact, demand, and resources, while maintaining clarity and brevity. The summary contains hashtags that align with the original post's tone and audience.]

Screenshot of output: Embrace the Future of AI Engineering: Innovate, Collaborate, and Lead

At the heart of modern technology lies AI Engineering, a dynamic field that bridges the gap between cutting-edge innovation and real-world impact. As AI continues to reshape industries, from healthcare to finance, AI Engineers are the architects of this transformation—fostering solutions that drive efficiency, accuracy, and growth.

What Makes AI Engineering Unique?

AI Engineering is more than coding; it's about building robust systems that can learn, adapt, and scale. Key skills include:

- Machine Learning (algorithm design, model tuning)
- Data Engineering (cleaning, processing, and integration)
- Cloud Computing (AWS, GCP, Azure)
- Collaboration (with data scientists, product managers, and domain experts)

Impact in Action

From predictive analytics to autonomous systems, AI Engineers are enabling breakthroughs:

- Healthcare: Personalized treatment plans, medical imaging analysis
- Finance: Fraud detection, algorithmic trading, risk management

- Manufacturing: Predictive maintenance, quality control
- Climate Tech: Climate modeling, energy optimization

☀️ Why Join the Movement?

The demand for AI Engineers is soaring—growth is exponential. As AI becomes a core competency, professionals who master this field will lead the charge in innovation.

🧩 Your Role in the Future

Whether you're a developer, data scientist, or product leader, AI Engineering offers a unique opportunity to:

- Shape the future of technology
- Drive impactful solutions
- Collaborate across disciplines

📚 Resources to Explore

- Books: *AI Ethics* by Yuval Noah Harari, *The AI Age* by Kai-Fu Lee
- Courses: Coursera's AI Engineering Specialization, Udacity's AI Nanodegree
- Communities: Join AI Engineering forums, LinkedIn groups, or GitHub projects

#AIEngineering #AIRevolution #TechInnovation #DataScience #FutureOfTech

Prompt Version 2 — Few-shot

Prompt Used: [Example 1:

"AI is transforming industries rapidly..."

Example 2:

“Learning AI engineering step-by-step helps build future-ready skills...”

Now write a similar LinkedIn post about AI Engineering.]

Output Received: [The model produced a more polished and professional post with better tone and structure.

AI Engineering: Shaping the Future of Technology

AI is revolutionizing industries by enabling smarter solutions, from healthcare to climate tech. AI Engineers are designing systems that learn, adapt, and scale, driving innovation across sectors. Key skills include machine learning, data engineering, cloud computing, and collaboration. The demand for AI Engineers is exploding, with opportunities to lead impactful projects and drive technological progress.

Resources like courses, books, and communities support continuous learning. Whether you're a developer, data scientist, or product leader, AI Engineering equips you to shape the future of technology.

🌟 #AIEngineering #TechInnovation #FutureOfTech #DataScience #AIRevolution.]

Observation: [

- Better writing style
- More human-like
- Improved engagement
- Examples helped guide output quality

AI Engineering is becoming one of the most important and rapidly growing fields in modern technology. Many industries such as healthcare, finance, manufacturing, and climate technology are using AI to solve complex problems and improve efficiency. AI Engineers require a combination of skills including machine learning, data engineering, cloud computing, and teamwork, showing that the field involves much more than just coding. AI systems are designed to learn, improve, and scale automatically, making them highly valuable for businesses and organizations. As companies continue adopting AI technologies, the demand for skilled AI Engineers is increasing rapidly, creating excellent career opportunities and leadership roles in the tech industry. The field also requires continuous learning because AI technologies evolve very quickly. Online platforms, books, and professional communities help learners build both theoretical and practical knowledge. In addition, collaboration through platforms like GitHub and LinkedIn plays an important role in career growth and knowledge sharing. Overall, AI

Engineering is not only focused on building advanced technologies but also on developing ethical and responsible AI solutions for the future.

]

Screenshot of output:



AI Engineering is Redefining the Future of Technology

AI Engineering is driving the next wave of innovation, empowering industries to tackle complex challenges with smarter solutions. From healthcare to finance, from manufacturing to climate tech, AI Engineers are building systems that learn, adapt, and scale—transforming how we live, work, and innovate.

At the heart of this revolution are key skills: machine learning, data engineering, cloud computing, and collaboration. Whether you're designing algorithms, cleaning data, or integrating systems, AI Engineering equips professionals to shape the future of technology.

Why Learn It?

The demand for AI Engineers is exploding. With AI becoming a core competency, mastering this field means unlocking opportunities to lead innovation, drive impact, and collaborate across disciplines.

Resources to Fuel Your Growth

- Courses: Coursera, Udacity
- Books: *AI Ethics*, *The AI Age*
- Communities: LinkedIn groups, GitHub projects

🌟 #AIEngineering #TechInnovation #FutureOfTech #DataScience #AIRevolution

Prompt Version 3 — Structured Prompt

Prompt Used: [Write a professional LinkedIn post about AI Engineering.]

Requirements:

- Start with a strong hook
- Mention AI tools
- Mention career growth
- Keep it under 150 words
- Add hashtags

]

Output Received: [The output was well-structured with a hook, concise explanation, and hashtags.

AI Engineering is the backbone of innovation, driving smarter solutions across industries.

From machine learning to data engineering, AI tools empower professionals to build scalable systems that transform how we live and work. As demand for skilled AI Engineers grows, this field offers unparalleled career growth, innovation, and impact. Mastering AI tools equips you to lead in a tech-driven world.

#AIEngineering #TechInnovation #FutureOfTech #CareerGrowth #DataScience]

Observation: [

- Best readability
- Followed instructions correctly
- Clear structure
- Most professional-looking output

I observed the following key elements in the original content:

1. Strong hook: The post starts with "🤖 AI Engineering is the backbone of innovation..." to grab attention.
2. AI tools: Emphasis on machine learning, data engineering, and cloud computing as core skills.
3. Career growth: Highlighted the rising demand for AI Engineers and the opportunities for professional development.
4. Call to action: The phrase "Mastering AI tools equips you to lead in a tech-driven

world" encourages proactive learning.

5. Hashtags: The post includes relevant hashtags like #AIEngineering, #TechInnovation, #FutureOfTech, #CareerGrowth, and #DataScience to align with LinkedIn's audience.]

Screenshot of output:

 AI Engineering is the backbone of innovation, driving smarter solutions across industries.

From machine learning to data engineering, AI tools empower professionals to build scalable systems that transform how we live and work. As demand for skilled AI Engineers grows, this field offers unparalleled career growth, innovation, and impact. Mastering AI tools equips you to lead in a tech-driven world.

#AIEngineering #TechInnovation #FutureOfTech #CareerGrowth #DataScience

Prompt Version 4 — JSON Prompt

Prompt Used: [Generate the response in JSON format:

```
{  
  "title": "",  
  "content": "",  
  "hashtags": []  
}
```

Topic: AI Engineering LinkedIn Post]

Output Received: [The model generated structured JSON containing title, content, and hashtags.

The JSON response captures a LinkedIn post about AI Engineering, emphasizing its role in driving innovation across industries, key tools (machine learning, data engineering), career growth, and impact. It includes relevant hashtags to align with the topic. The summary retains the core message while maintaining the relevance.

]

Observation: [

- Good for automation
- Structure mostly correct

- Slight formatting inconsistency occasionally

The JSON response reflects a LinkedIn post about AI Engineering, emphasizing its role in driving innovation across industries. The title highlights its importance in the tech era. The content begins with a strong hook (robot emoji) to engage readers, highlights key skills (machine learning, data engineering, cloud computing), and underscores the rising demand for AI Engineers, career growth, and impact. It also emphasizes the value of mastering AI tools to lead in a tech-driven world. The hashtags align with the post's theme, reinforcing relevance to the audience. The structure balances technical details (skills, tools) with professional growth and impact, making it both informative and motivational.

]

Screenshot of output:

```
{ "title": "AI Engineering: Driving Innovation in the Tech Era",  
  "content": "🤖 AI Engineering is the backbone of innovation, driving smarter solutions across industries. From machine learning to data engineering, AI tools empower professionals to build scalable systems that transform how we live and work. As demand for skilled AI Engineers grows, this field offers unparalleled career growth, innovation, and impact. Mastering AI tools equips you to lead in a tech-driven world.",  
  "hashtags": ["#AIEngineering", "#TechInnovation", "#FutureOfTech", "#CareerGrowth", "#DataScience"] }
```

7. Output Comparison

Prompt Style	Was the output clear?	Did the model follow instructions?	Was the output useful?	Notes
Zero-shot	[Medium]	[Partial]	[Medium]	[Generic response]
Few-shot	[Good]	[Good]	[Good]	[Better tone & style]
Structured	[Excellent]	[Excellent]	[Excellent]	[Best overall result.]

Prompt Style	Was the output clear?	Did the model follow instructions?	Was the output useful?	Notes
JSON	[Good]	[Mostly]	[Good]	[Useful for APIs/ Automation]

8. Performance Notes

- **How fast did the model respond overall?:** [The model responded within a few seconds for short prompts.]
- **Did the model feel slow on your system?:** [No major slowdown was observed for normal text generation tasks.]
- **Did any prompt type work noticeably better?:** [Yes. Structured prompts produced the best and most accurate outputs.]
- **Did the model struggle with structure or formatting?:** [The JSON prompt occasionally produced small formatting inconsistencies.]
- **Did you observe hallucination or weak answers?:** [Yes. The zero-shot prompt produced generic.]

Screenshot showing one strong output:

Structured:  AI Engineering is the backbone of innovation, driving smarter solutions across industries.

From machine learning to data engineering, AI tools empower professionals to build scalable systems that transform how we live and work. As demand for skilled AI Engineers grows, this field offers unparalleled career growth, innovation, and impact. Mastering AI tools equips you to lead in a tech-driven world.

#AIEngineering #TechInnovation #FutureOfTech #CareerGrowth #DataScience

Screenshot showing one weak / incorrect / less useful output:

Zero shot: Embrace the Future of AI Engineering: Innovate, Collaborate, and Lead

At the heart of modern technology lies AI Engineering, a dynamic field that bridges the

gap between cutting-edge innovation and real-world impact. As AI continues to reshape industries, from healthcare to finance, AI Engineers are the architects of this transformation—fostering solutions that drive efficiency, accuracy, and growth.

What Makes AI Engineering Unique?

AI Engineering is more than coding; it's about building robust systems that can learn, adapt, and scale. Key skills include:

- Machine Learning (algorithm design, model tuning)
- Data Engineering (cleaning, processing, and integration)
- Cloud Computing (AWS, GCP, Azure)
- Collaboration (with data scientists, product managers, and domain experts)

Impact in Action

From predictive analytics to autonomous systems, AI Engineers are enabling breakthroughs:

- Healthcare: Personalized treatment plans, medical imaging analysis
- Finance: Fraud detection, algorithmic trading, risk management
- Manufacturing: Predictive maintenance, quality control
- Climate Tech: Climate modeling, energy optimization

Why Join the Movement?

The demand for AI Engineers is soaring—growth is exponential. As AI becomes a core competency, professionals who master this field will lead the charge in innovation.

Your Role in the Future

Whether you're a developer, data scientist, or product leader, AI Engineering offers a unique opportunity to:

- Shape the future of technology
- Drive impactful solutions
- Collaborate across disciplines

Resources to Explore

- Books: *AI Ethics* by Yuval Noah Harari, *The AI Age* by Kai-Fu Lee
- Courses: Coursera's AI Engineering Specialization, Udacity's AI Nanodegree
- Communities: Join AI Engineering forums, LinkedIn groups, or GitHub projects

#AIEngineering #AIRevolution #TechInnovation #DataScience #FutureOfTech

9. Final Reflection

- **Best prompt style for this local model:** [Structured Prompting]
- **Weakest prompt style for this local model:** [Zero Shot Prompting]
- **What worked well in local LLM usage?:** [
 - ❖ Fast local inference
 - ❖ No internet dependency
 - ❖ Good privacy
 - ❖ Strong prompt experimentation capability]
- **What limitations did you observe?:** [
 - ❖ Limited creativity compared to cloud AI models
 - ❖ Slight formatting errors in structured outputs
 - ❖ Performance depends heavily on hardware specifications.]
- **In what kind of real-world work would you use this local model?:** [

- ❖ Content writing
- ❖ Draft generation
- ❖ Coding assistance
- ❖ Learning AI concepts
- ❖ Social media post generation.]
- **Would you recommend local LLMs for beginners? Why or why not?:** [Yes. Local LLMs are excellent for learning AI concepts, prompt engineering, and private experimentation without API costs.]

10. Submission Checklist

Added system details

- ☒ Added system details
- ☒ Installed and ran Ollama
- ☒ Selected one local model
- ☒ Choose one real-world task
- ☒ Tested 4 prompt styles
- ☒ Added outputs
- ☒ Added comparison table
- ☒ Added screenshots
- ☒ Added final reflection

Make sure your screenshots are clear and readable. Screenshots are required as proof of setup, prompt testing, and output observation.